

Evaluation of Multilingual Ability to Use Spatial Deictic Expressions in Vision-Language Models

视觉语言模型中多语言使用空间指示表达能力的评估

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Abstract

摘要

One of the expected abilities of vision-language models (VLMs) is spatial reasoning ability based on a given text and image. To evaluate the spatial reasoning abilities of VLMs, we focus on the use of spatial deictic expressions, which are defined as spatial expressions whose referent is determined by their situational context, such as “this” and “that”. To handle spatial deictic expressions, VLMs must jointly reason over language and visual space, grounding context-dependent references in the image’s spatial structure. In addition, selecting appropriate spatial deictic expressions across languages requires VLMs to understand the language-specific spatial distinctions encoded by these expressions. In this paper, we develop a benchmark¹ to evaluate the multilingual ability of VLMs to use spatial deictic expressions in four languages. Our experiments using this benchmark reveal that the tested models use demonstratives in a manner different from that of humans, particularly in selecting the appropriate demonstratives based on the distance to the object.

视觉语言模型（VLMs）的预期能力之一是基于给定文本和图像的空间推理能力。为了评估 VLMs 的空间推理能力，我们关注空间指示表达式的使用，这些表达式被定义为其指代对象由情境上下文决定的空间表达式，例如“this”和“that”。为了处理空间指示表达式，VLMs 必须联合地在语言和视觉空间上进行推理，将情境依赖的指代锚定在图像的空间结构中。此外，选择适当的跨语言空间指示表达式需要 VLMs 理解这些表达式所编码的语言特定空间差异。在本文中，我们开发了一个基准¹，用于评估 VLMs 在四种语言中使用空间指示表达式的多语言能力。使用该基准进行的实验表明，被测试的模型在使用指示代词方面与人类有所不同，特别是在根据与物体的距离选择适当的指示代词方面。

1 Introduction

1 引言

In recent years, large language models (LLMs) and vision-language models (VLMs) have achieved remarkable progress due to their scalability. Furthermore, a key feature of LLMs and VLMs is their ability to handle a wide range of tasks not only in English but also in multiple languages. The development prompted vigorous attempts to evaluate their reasoning ability. Previous studies have investigated the abilities of VLMs to capture spatial relations and spatial expressions, such as frames of reference² (Zhang et al., 2025b; Khemlani et al., 2025).

近年来，由于其可扩展性，大型语言模型（LLMs）和视觉-语言模型（VLMs）在各方面都取得了显著进展。此外，LLMs 和 VLMs 的一个关键特性是它们能够处理不仅限于英语，还包括多种语言的各种任务。

这种发展促使人们积极尝试评估它们的推理能力。先前的研究已经探讨了 VLMs 捕捉空间关系和空间表达的能力，例如参照系² (Zhang 等, 2025b; Khemlani 等, 2025)。

Despite efforts, the ability of VLMs to utilize spatial deictic expressions, an important type of spatial expression, has not been explicitly studied. Deixis is the usage of expressions whose referent is dependent on the situation of utterance. Spatial deictic expressions are expressions that depend on the space in which the utterance occurred, such as “here” or “that”. For example, suppose there is a pen in front of the speaker. If the pen is far away, especially at an unreachable point, the speaker tends to say “that pen”, while if it is near the speaker, the speaker may indicate it by “this pen”. As this example shows, deictic functions are fundamental to human language because they connect linguistic expressions with the physical environment. Therefore, the evaluation of a VLM’s proficiency in using spatial deictic expressions has linguistic significance for benchmarking its capacity for spatial reasoning.

尽管有诸多努力，VLMs（视觉语言模型）在利用空间指示表达（一种重要的空间表达类型）方面的能力尚未被明确研究。指示（deixis）是指那些指代依赖于话语情境的表达方式。空间指示表达是指依赖于话语发生空间的表达方式，例如“这里”或“那里”。例如，假设说话者面前有一支笔。如果这支笔离说话者很远，特别是位于无法到达的位置，说话者倾向于说“那支笔”；而如果它离说话者很近，说话者可能会用“这支笔”来指示它。正如这个例子所示，指示功能对于人类语言来说是基本的，因为它们将语言表达与物理环境联系起来。因此，评估 VLM 在使用空间指示表达方面的熟练程度，对于基准测试其空间推理能力具有语言学意义。

We argue that the use of spatial deixis poses two major challenges for VLMs: cross-linguistic variation and inherent ambiguity. First, to use spatial deictic expressions appropriately, multilingual VLMs need to understand their semantic differences across languages. For example, while in English we use two demonstratives, proximal³ “this” and distal “that”, in Japanese we use three demonstratives, proximal kono, distal ano, and medial sono. As such, across languages, the number of kinds of spatial deictic expressions varies. In addition, as in another example, although Spanish and Japanese both have three demonstratives: proximal, distal, and medial, the meaning of each word is not the same. According to Diessel (1999), Spanish medial ese signifies a relatively intermediate place between the speaker and the addressee, while Japanese medial sono indicates a place near the addressee. Therefore, multilingual VLMs need

我们认为，空间指示的使用对视觉语言模型（VLMs）提出了两个主要挑战：跨语言变异性和内在歧义性。首先，为了恰当地使用空间指示表达，多语言 VLMs 需要理解其不同语言中的语义差异。例如，虽然在英语中我们使用两个指示代词，近指³ “this” 和远指 “that”，但在日语中我们使用三个指示代词，近指 kono，远指 ano，以及中指 sono。因此，跨语言来看，空间指示表达的种类数量存在差异。此外，如另一个例子所示，虽然西班牙语和日语都有三个指示代词：近指、远指和中指，但每个词的含义并不相同。根据 Diessel (1999) 的观点，西班牙语的中指 ese 表示说话者与听话者之间的相对中间位置，而日语的中指 sono 表示靠近听话者的位置。因此，多语言 VLMs 需要

to capture such subtle differences in the semantic nuances of spatial distances across languages to employ demonstratives in a manner similar to humans. Second, Coventry et al. (2023), a study investigating demonstrative usage among approximately 30 participants per language, reported that individuals do not necessarily use the same demonstrative in the same situation; there are individual differences in capturing the semantic nuances of spatial distance. Thus, if multilingual VLMs succeed in learning spatial deixis expressions, they are expected to recognize differences across languages and reproduce the distribution of human performance in the use of demonstratives.

为了捕捉语言中空间距离语义细微差异的微妙之处，并以类似于人类的方式使用指示代词。其次，Coventry 等人 (2023) 的一项研究调查了每种语言大约 30 名参与者在指示代词使用方面的行为，报告指出，人们在相同的情境下并不一定使用相同的指示代词；在捕捉空间距离的语义细微差异方面存在个体差异。因此，如果多语言视觉语言模型能够成功学习空间指示表达方式，那么它们有望识别不同语言之间的差异，并再现人类在使用指示代词时的表现分布。

Based on these points, we developed a benchmark to measure the ability to use spatial deictic expressions in various languages, such as English and Japanese, and evaluated VLMs. The task

included in the benchmark is designed to investigate how the absolute distance of an object influences the choice of demonstratives of VLMs. We also analyzed the performance of VLMs by comparing the results of experiments on human spatial deictic expressions reported in Coventry et al. (2023).

基于这些要点，我们开发了一个基准测试，用于衡量在各种语言（如英语和日语）中使用空间指示表达的能力，并评估视觉语言模型（VLMs）。该基准测试中的任务旨在研究物体的绝对距离如何影响 VLMs 选择指示代词。我们还通过比较 Coventry 等人（2023）报告的人类空间指示表达实验结果，分析了 VLMs 的性能。

The contributions of this paper are as follows:

本文的贡献如下：

1. We constructed the first benchmark to evaluate the ability to employ demonstratives in VLMs.
2. We revealed that open VLMs fail to use demonstratives in a human-like way, and the differences between humans vary across models. In particular, VLMs do not show the shift in the selection of demonstratives with distance, as observed in humans.
3. 我们构建了第一个基准，用于评估 VLMs 在使用指示代词方面的能力。
4. 我们发现开放的 VLMs 无法以类似人类的方式使用指示代词，且不同模型之间的人类差异存在差异。特别是，VLMs 在距离变化时，不会像人类那样表现出指示代词选择的转变。

2 Background

2 背景

2.1 Analysis of Spatial Reasoning Ability in VLMs

2.1 对 VLMs 空间推理能力的分析

A large number of evaluations have been conducted on spatial reasoning ability in VLMs (Liu et al., 2025). Those previous benchmarking works revealed that VLMs still lack sufficient human-level ability (Khemlani et al., 2025; Zhang et al., 2025a), especially for 3D world spatial reasoning tasks such as distance estimation (Yang et al., 2025; Zhang et al., 2025a) and positional relationships (e.g. front or left) (Liu et al., 2023; Khemlani et al., 2025).

在 VLMs 中对空间推理能力进行了大量评估（Liu 等，2025）。这些先前的基准测试工作表明，VLMs 仍然缺乏足够的与人类相当的能力（Khemlani 等，2025；Zhang 等，2025a），尤其是在 3D 世界空间推理任务方面，例如距离估计（Yang 等，2025；Zhang 等，2025a）和位置关系（如前或左）（Liu 等，2023；Khemlani 等，2025）。

Moreover, many benchmarks were developed only in English (the works cited in the preceding paragraph are all for English evaluation). Only a few benchmarks contain tasks to evaluate the multilingual ability of spatial reasoning in VLMs (Liu et al., 2024; Haller et al., 2025; Zhang et al., 2025b).

此外，许多基准测试仅以英语开发（上一段引用的文献均为针对英语的评估）。只有少数基准测试包含用于评估视觉语言模型（VLMs）空间推理多语言能力的任务（Liu 等，2024；Haller 等，2025；Zhang 等，2025b）。

Liu et al. (2024) aims to evaluate the ability of VLMs robustly and holistically. To realize this goal, the benchmark, MMBench, includes a task called “spatial relationship” that contains 2D spatial relationships, and “physical relation” that contains 3D spatial relationships. The authors created a Chinese version of the benchmark, called MMBench-CN. They demonstrated that most of the tested VLMs performed worse on MMBench-CN, but the gap in performance between English and Chinese for a model that achieves a high English score may be smaller.

刘等人 (2024) 旨在评估视觉语言模型 (VLMs) 在稳健性和全面性方面的能力。为了实现这一目标, 基准测试集 MMBench 包含了一个名为 “空间关系” 的任务, 涵盖了二维空间关系, 以及一个名为 “物理关系” 的任务, 涵盖了三维空间关系。作者创建了该基准测试的中文版本, 称为 MMBench-CN。他们表明, 大多数测试的 VLMs 在 MMBench-CN 上的表现较差, 但对于在英文方面取得高分的模型, 其英文和中文之间的性能差距可能会更小。

Haller et al. (2025) constructed a benchmark called PISA-BENCH by collecting questions from the PISA test, an international assessment of the academic performance of students. It contains “spatial and geometric reasoning” tasks. As the name implies, it contains a task that asks for the shape of an object viewed from behind. As a result, they found a significant gap in performance between English and the others.

Haller 等人 (2025) 通过收集 PISA 测试中的问题, 构建了一个名为 PISA-BENCH 的基准测试, PISA 是一项国际性的学生学业表现评估。它包含了 “空间与几何推理” 任务。正如其名称所示, 它包含一个任务, 要求判断从物体后方观察时的形状。因此, 他们发现英语与其他语言之间存在显著的性能差距。

Zhang et al. (2025b) developed an evaluation protocol called COMFORT. The authors focused on frames of reference (FoR) to evaluate the robustness, consistency, and flexibility of VLMs’ spatial reasoning. They constructed images of datasets using Blender (Blender Online Community, 2016). The benchmark asks VLMs to answer whether one object has an indicated relationship to the other, for example, “From the camera’s viewpoint, is the ball behind the car?” using an image of a ball and a car. They concluded that VLMs lack the ability to use indicated FoR and have a bias to the English method of expression rather than a method specific to the language.

张等人 (2025b) 开发了一个名为 COMFORT 的评估协议。作者们专注于参考系 (FoR) 来评估视觉语言模型 (VLMs) 的空间推理能力的鲁棒性、一致性和灵活性。他们使用 Blender (Blender Online Community, 2016) 构建了数据集的图像。基准测试要求 VLMs 回答一个物体与另一个物体之间是否存在指示的关系, 例如, “从摄像机的视角看, 球是否在车的后面?” 使用一张球和一辆车的图像。他们得出结论, VLMs 缺乏使用指示参考系的能力, 并且更倾向于英语表达方式, 而不是特定于语言的表达方法。

However, previous work does not focus on the extent to which VLMs handle spatial deictic expressions in spatial reasoning tasks. In this paper, we analyze multilingual abilities to use spatial deictic expressions in VLMs.

然而, 以往的研究并未关注 VLMs 在空间推理任务中处理空间指示表达的程度。在本文中, 我们分析了 VLMs 使用空间指示表达的多语言能力。

2.2 Spatial Deixis

2.2 空间指涉

2.2.1 Deixis and Demonstratives

2.2.1 指示词与指示代词

Deixis is the usage of spatial expressions whose referent is dependent on the situation of utterance (Saito et al., 2015). Deictic expressions are actual expressions of deixis. For example, given

指涉 (Deixis) 是指使用空间表达方式, 其指代对象依赖于话语情境 (Saito 等, 2015)。指示表达是指示

的实质表达。例如，给定

the sentence John is here., the word here is a spatial deictic expression, so we cannot decide what the word here refers to without considering the situational context of the utterance. In this situation, the speaker doesn't need to explain the referent of Here. Instead, the listener has to determine the referent of here. In this way, the accurate recognition of the meaning of spatial deictic expressions requires precisely determining the referent of the word.

句子“John is here.”中的“here”是一个空间指示表达，因此在不考虑话语所处的情境背景的情况下，我们无法确定“here”指的是什么。在这种情况下，说话人不需要解释“here”的指代对象，而是听者必须确定“here”的指代对象。通过这种方式，准确识别空间指示表达的含义，需要精确地确定词语的指代对象。

Deictic expressions are classified into several classes (Diessel, 1999; Fillmore, 1997), and spatial deictic expressions are one of them. Spatial deictic expressions are deictic expressions whose referent is related to space. In this paper, we focus on spatial deictic expressions, especially demonstratives.

指示表达被分为几类 (Diessel, 1999; Fillmore, 1997)，空间指示表达是其中之一。空间指示表达是指其指称对象与空间有关的指示表达。在本文中，我们关注空间指示表达，特别是指示代词。

Coventry et al. (2023) investigated how people with various native languages use different demonstratives. In their experiments, most languages have two or three demonstratives. The differences among demonstratives vary across languages, but, in general, they tend to depend on distance. Especially, a demonstrative used to refer to nearby and faraway objects from the speaker is called proximal and distal, respectively.

Coventry 等人 (2023) 研究了不同母语者如何使用不同的指示代词。在他们的实验中，大多数语言都有两个或三个指示代词。指示代词之间的差异在不同语言中有所不同，但总体而言，它们往往取决于距离。特别是，用来指代说话者附近和远处物体的指示代词分别称为近指代词和远指代词。

Some languages also have another criterion to differentiate demonstratives. For example, in Japanese language, ano and sono both refer to the object not near the speaker, but the difference depends on the distance from the addressee. Coventry et al. (2023) revealed that this kind of dependency on distance from the addressee is also observed in some languages, such as Finnish or Korean.

一些语言还具有另一个标准来区分指示代词。例如，在日语中，ano 和 sono 都指说话者附近没有的物体，但差异取决于听话者的位置距离。Coventry 等人 (2023) 发现，这种依赖于听话者距离的特征在一些语言中也存在，例如芬兰语或韩语。

In order to output spatial deictic expressions appropriately, multilingual VLMs have to recognize the differences in demonstratives among languages.

为了适当地输出空间指示表达，多语言 VLMs 必须识别不同语言中指示代词的差异。

2.2.2 Deixis and Language Models

2.2.2 指示语和语言模型

To the best of our knowledge, whether VLMs can handle spatial deictic expressions has not been sufficiently explored. Existing attempts to evaluate the deixis understanding ability of language models are prone to focus on discourse deixis, a kind of deixis, because LLMs do not have access to external visual contextual information. A benchmark called PUB (Sravanthi et al., 2024) evaluates the pragmatic reasoning abilities of LLMs, including understanding of deixis. Their results show that almost all tested LLMs perform worse than humans, and the authors conclude that the poor scores are not due to world knowledge but to a lack of pragmatic reasoning. While several studies of deixis and language models exist in the field of humancomputer interaction (Lai et al., 2025;

Han and Isaacs, 2025), there are few studies of evaluation in vision-language models.

据我们所知，目前对视觉语言模型（VLMs）是否能够处理空间指示表达的研究尚不充分。现有的评估语言模型指示理解能力的尝试往往侧重于话语指示，这是一种指示类型，因为大语言模型（LLMs）无法获取外部的视觉上下文信息。一个名为 PUB 的基准测试（Sravanthi 等，2024）评估了大语言模型的语用推理能力，包括对指示的理解。他们的结果显示，几乎所有测试的大语言模型的表现都不如人类，作者得出结论，低分并非由于世界知识的缺乏，而是由于缺乏语用推理能力。尽管在人机交互领域已有几项关于指示和语言模型的研究（Lai 等，2025；Han 和 Isaacs，2025），但针对视觉语言模型的评估研究却很少。

3 Method

3 方法

To construct our benchmark, we refer to the linguistic experiments called “memory game” (Gudde et al., 2018). A detailed description of the “memory game” is provided as follows.

为了构建我们的基准测试，我们参考了被称为“记忆游戏”（Gudde 等，2018）的语言实验。对“记忆游戏”的详细描述如下。

3.1 Memory Game

3.1 记忆游戏

The memory game is a method to investigate the usage of demonstratives of humans without subjects noticing that linguistic investigation is conducted (Gudde et al., 2018)⁴. This method aims to elicit demonstratives from subjects of the experiment in experimentally controllable and naturalistic situations. To ensure the naturalistic setting, the experiment is presented as a memory experiment, and the purpose of investigating demonstrative usage is concealed from participants until the experiment is completed.

记忆游戏是一种方法，用于研究人类在未察觉语言研究正在进行的情况下使用指示代词的情况（Gudde 等，2018）⁴。该方法旨在实验可控且自然的情境下，从实验对象那里引出指示代词的使用。为确保自然情境，实验被呈现为一个记忆实验，并且直到实验结束，才向参与者隐瞒研究指示代词使用的目的。

The procedure of the memory game in Gudde et al. (2018) is as follows. First, subjects sit on one side of a long desk and are told that the experiments examine the influence of language on memory. Then they are instructed to name the shape drawn on the disk, which is placed on the long desk, by pointing their finger at the disk. When naming the object, the subjects must use three words: a demonstrative, a color, and the shape of the object, like “that black cross”. Experimenters record which demonstrative is used, and one trial of the experiment is done. Experiments are conducted multiple times with changes in some variables, such as the disk’s location. The places of the disk are classified into three “regions” by distance, region 1 for 25-75 cm, region 2 for 100-150 cm, and region 3 for 175-225 cm. The result of the experiment showed that these parameters have an influence on demonstrative use.

Gudde 等人（2018）的回忆游戏程序如下。首先，被试者坐在长桌的一侧，并被告知实验将考察语言对记忆的影响。然后，他们被指示通过指向盘子来命名放置在长桌上的图形。在命名物体时，被试者必须使用三个词：指示代词、颜色和物体的形状，例如“那个黑色十字”。实验者记录使用了哪个指示代词，完成一次实验。实验会多次进行，某些变量会发生变化，例如盘子的位置。盘子的位置根据距离分为三个“区域”：区域 1 为 25-75 厘米，区域 2 为 100-150 厘米，区域 3 为 175-225 厘米。实验结果表明，这些参数对指示代词的使用有影响。

Coventry et al. (2023) conducted a memory game across various languages, and they analyzed how the usages of demonstratives are different among languages. In their research, experiments were conducted with 29 languages (e.g., Japanese and English) and 874 subjects whose native

language was one of them. The memory game experiment is conducted by using a white disk with a

Coventry 等人 (2023) 在多种语言中进行了一项记忆游戏实验, 他们分析了指示代词在不同语言中的使用差异。在他们的研究中, 对 29 种语言 (例如日语和英语) 以及 874 名母语为其中一种语言的受试者进行了实验。记忆游戏实验是通过使用一个白色圆盘来进行的。

shape drawn on it. The variables are places, shapes, colors, and the addressee’s location. Places of the disk are classified into three “regions” by distance, like Gudde et al. (2018), but they set regions in a slightly different way: region 1 for 25-50 cm, region 2 for 150-175 cm, and region 3 for 275-300 cm. Shape is a shape drawn on the disk (e.g., a star, a cross, and a triangle). Color is the color of the shape. Through extensive experiments, they showed that native speakers use different demonstratives (distal, medial, and proximal) depending on the distance to the target. Furthermore, they found that the usage patterns of these demonstratives across distances vary across languages; for example, Japanese speakers tend to use medial demonstratives (sono) for objects at an intermediate distance, whereas Korean speakers use distal ones (jeo).

在上面绘制的形状。变量包括地点、形状、颜色以及收件人的位置。磁盘上的地点根据距离被分为三个“区域”, 类似于 Gudde 等人 (2018) 的研究, 但他们的区域划分方式略有不同: 区域 1 是 25-50 厘米, 区域 2 是 150-175 厘米, 区域 3 是 275-300 厘米。形状是指在磁盘上绘制的形状 (例如星星、十字形和三角形)。颜色是形状的颜色。通过大量实验, 他们表明母语者会根据目标的距离使用不同的指示代词 (远指、中指和近指)。此外, 他们发现这些指示代词在不同距离上的使用模式在语言之间有所不同; 例如, 日语使用者倾向于在中等距离的对象上使用中指代词 (sono), 而韩语使用者则使用远指代词 (jeo)。

The memory game enables the evaluation of human demonstrative use across multiple languages, accounting for subtle ambiguities that arise depending on the distance between the speaker and the object. In this study, we construct a benchmark for evaluating demonstrative use in VLMs based on the memory game paradigm.

记忆游戏使得评估人类在多种语言中的指示代词使用成为可能, 考虑到了说话者与所指对象之间的距离所带来的细微歧义。在本研究中, 我们基于记忆游戏范式构建了一个基准, 用于评估视觉语言模型中的指示代词使用。

3.2 Task Setting

3.2 任务设置

Based on the memory game, we composed a VQA task to analyze the extent to which VLMs capture spatial deixis across languages. We chose four languages for evaluation, Japanese, Korean, English, and Chinese, in which there were differences with respect to the usage of spatial deictic expressions. There exist three demonstratives in Japanese (proximal kono, distal ano and medial sono) and Korean (proximal i, distal jeo and medial geu), and two demonstratives in English (proximal this, and distal that) and Chinese (proximal zhè ge, and distal nà ge).

基于记忆游戏, 我们设计了一个 VQA 任务, 以分析视觉语言模型 (VLMs) 在跨语言中捕捉空间指示 (spatial deixis) 的程度。我们选择了四种语言进行评估, 即日语、韩语、英语和汉语, 这些语言在空间指示表达的使用上存在差异。日语中有三种指示代词 (近指 kono、远指 ano 和中指 sono), 韩语中有三种指示代词 (近指 i、远指 jeo 和中指 geu), 英语中有两种指示代词 (近指 this 和远指 that), 汉语中有两种指示代词 (近指 zhè ge 和远指 nà ge)。

We follow the task setting used in the memory game. In each image (see Figure 1), one white disk with a colored shape is put on the black desk. VLMs are instructed to describe the shape drawn on the disk, using a fixed format consisting of a demonstrative, a color, and the shape, such as “this red circle.”

我们采用记忆游戏中使用的任务设置。在每张图像 (见图 1) 中, 一个带有彩色形状的白色圆盘被放置在黑色桌子上。VLMs 被指示使用由指示代词、颜色和形状组成的固定格式来描述圆盘上的形状, 例如 “这

个红色圆圈”。

3.3 Benchmark Construction

3.3 基准构建

We constructed the datasets with Blender (Blender Online Community, 2016) following the construction process of previous benchmarks (Khemlani et al., 2025; Zhang et al., 2025b), because we can

我们根据之前基准数据集的构建过程（Khemlani 等人，2025；Zhang 等人，2025b），使用 Blender（Blender 在线社区，2016）构建了这些数据集，因为我们可以

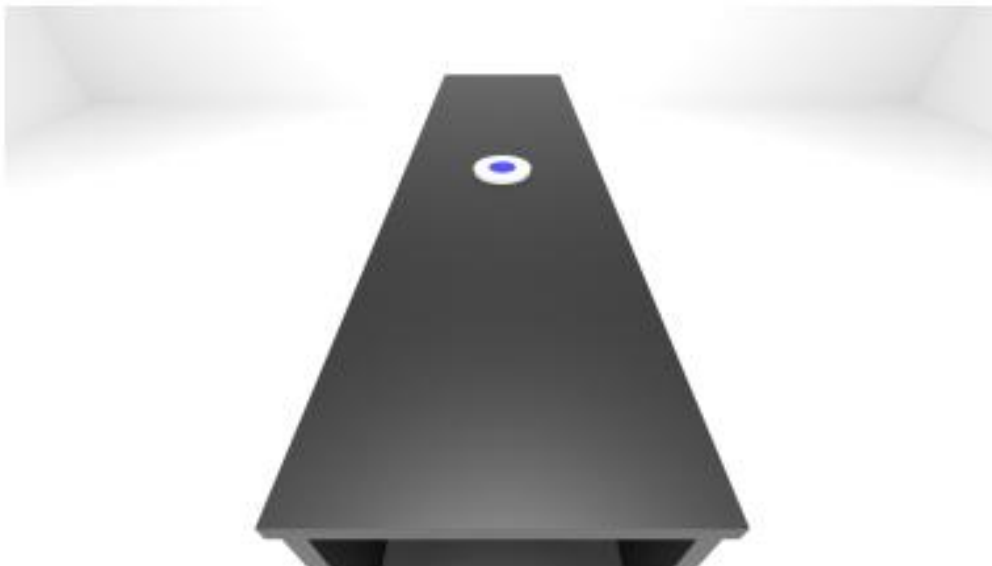


Figure 1: Examples of images included in the benchmark. Distance from the object is 0.25m, 1.50m, and 2.75m, from left to right.

图 1：基准集包含的图像示例。从左到右，物体的距离分别为 0.25 米、1.50 米和 2.75 米。

easily tweak the location of objects and the brightness with it. To be consistent with experimental settings regulated in Gudde et al. (2018), we produced images with the aligned settings. For example, we used a long desk without any patterns and did not place any objects around it, which can help to estimate the distance of the object. We also tweaked the camera height or the room brightness so that VLMs could obtain sufficient information to answer.

可以轻松调整物体的位置和亮度。为了与 Gudde 等人 (2018) 中规定的实验设置保持一致，我们使用对齐的设置生成图像。例如，我们使用了一张没有任何图案的长桌，并且没有在周围放置任何物体，这有助于估计物体的距离。我们还调整了相机高度或房间亮度，以便 VLMs 能够获取足够的信息来回答问题。

Examples of the image included in the benchmark are displayed in Figure 1. On the desk, we put a disk with a solid-colored shape. The disk was placed at one of the distances 0.25 m, 1.50 m, and 2.75 m from the subject, as displayed in Figure 1, corresponding to region 1, 2, and 3 in Coventry et al. (2023) respectively. We used 5 figures (circle, cross, square, star, and triangle) and 4 colors (black, blue, green, and red) to create 20 images per region, resulting in 60 images in total (5 shapes times 4 colors for the shape $\times$ 3 positions for the disk).

基准中包含的图像示例如图 1 所示。在桌子上，我们放置了一个具有单一颜色形状的圆盘。该圆盘放置在被试者前方 0.25 米、1.50 米和 2.75 米三个距离之一处，如图 1 所示，分别对应于 Coventry 等人 (2023) 中的区域 1、2 和 3。我们使用了 5 种图形（圆形、十字形、正方形、星形和三角形）和 4 种颜色（黑色、蓝色、绿色和红色）来为每个区域创建 20 张图像，总共生成了 60 张图像（图形 $\times$ 颜色组合为 5 种形状 $\times$ 4 种颜色，圆盘位置为 3 种）。

4 Experimental Setting

4 实验设置

4.1 Model

4.1 模型

We chose Gemma 3 4B⁵ and Gemma 3 12B⁶ from Gemma 3 series (Gemma Team, 2025), and Qwen3-

我们从 Gemma 3 系列 (Gemma 团队, 2025) 中选择了 Gemma 3 4B⁵ 和 Gemma 3 12B⁶，以及 Qwen3-

VL 8B⁷ and Qwen3-VL 32B⁸ from Qwen3-VL series (Bai et al., 2025) for evaluation. All of them are open, instruction-tuned, and multilingual VLMs. We avoided closed models such as GPT-5.2 because we cannot obtain raw logits for analysis from their APIs. We accessed trained models of them through Hugging Face Hub⁹. Output token decoding was performed using a greedy algorithm for all models.

VL 8B⁷ 和 Qwen3-VL 32B⁸ 来自 Qwen3-VL 系列 (Bai 等, 2025) 用于评估。它们都是开放的、指令调优的、多语言的视觉语言模型 (VLM)。我们避免使用闭源模型如 GPT-5.2，因为我们无法通过其 API 获取用于分析的原始 logit。我们通过 Hugging Face Hub⁹ 访问了这些模型的训练版本。所有模型均使用贪婪算法进行输出 token 解码。

4.2 Prompt

4.2 提示语

Since the original instructions used in Coventry et al. (2023) experiments are not publicly available, we design prompts for VLMs following the protocol described in Coventry et al. (2023). We configured a prompt to have VLMs output the same format as the memory game for humans: a demonstrative, the color of the object, and a shape drawn on the disk placed on the long desk. The prompt used in our experiments is shown below.

由于科文特等人（2023）实验中使用的原始指令并未公开，我们根据科文特等人（2023）描述的协议设计了用于视觉语言模型（VLMs）的提示。我们配置了一个提示，使 VLMs 输出与人类在记忆游戏中相同的格式：一个示范、物体的颜色以及放置在长桌上的盘子上绘制的形状。我们实验中使用的提示如下所示。

Prompt: Analyze the image and identify the shape on the disk. Describe it by filling in the following three-word template exactly:

提示：分析图像并识别磁盘上的形状。请按照以下三个词的模板准确描述它：

[Demonstrative] [Color] [Shape]

[指示] [颜色] [形状]

Constraints:

约束条件：

Use “This” or “That” for the demonstrative.

使用 “This” 或 “That” 来表示指示代词。

Use a single word for the color.

用一个单词表示颜色。

Use a single word for the shape.

用一个词表示形状。

Output only the three words. Do not include a period or any introductory text.

只输出三个词。不要包含句号或任何引导性文字。

This prompt was written in English at first, and then translated into each language by Google Gemini¹⁰. After the translation, we added lacking demonstratives, especially medial demonstratives of Japanese and Korean, because the literal translation of the prompt contained only proximal and distal demonstratives.

这个提示最初是用英文写的，然后由 Google Gemini 翻译成每种语言¹⁰。在翻译之后，我们添加了缺失的指示代词，尤其是日语和韩语中的中距离指示代词，因为提示的直译版本中只包含近指和远指指示代词。

4.3 Metrics

4.3 指标

We primarily employ two metrics. One is a probability distribution of the use of demonstratives for each VLM. The other is the Jensen-Shannon distance between the probability distributions of

demonstratives by models and humans. This al-

我们主要采用两个指标。一个是每个 VLM 使用指示代词的概率分布。另一个是模型和人类在指示代词概率分布上的 Jensen-Shannon 距离。这

allows us to analyze how precisely models reproduce human usage of the spatial deictic expressions.

让我们分析模型在再现人类使用空间指示表达方面的精确程度。

The probability distribution of the use of demonstratives is calculated from the logits obtained from outputs of LLMs. We first calculate the logits of all target demonstratives, and then normalize them by applying the softmax function. For example, we calculate the probability for “this” in English as $P(this)/(P(this) + P(that))$.

指示代词的使用概率分布是从大型语言模型（LLMs）输出的 logits 计算得出的。我们首先计算所有目标指示代词的 logits，然后通过应用 softmax 函数对它们进行归一化。例如，我们计算英语中 “this” 的概率为 $P(this)/(P(this) + P(that))$ 。

The fidelity of humans’ selection of demonstratives is calculated referring to the probability distribution of humans reported on Coventry et al. (2023). We regarded the results of Coventry et al. (2023), which were conducted for humans, as the probability distribution of demonstratives inherent in each language. We measured the Jensen-Shannon distance between the distributions of humans and VLMs. We adopted the Jensen-Shannon distance because it can measure the distance for a distribution that has a class whose probability is 0.

人类选择指示代词的忠实度是参照 Coventry 等人 (2023) 报告的人类概率分布来计算的。我们将 Coventry 等人 (2023) 针对人类进行的研究结果视为每种语言中指示代词固有的概率分布。我们测量了人类分布与 VLMs 分布之间的 Jensen-Shannon 距离。我们采用 Jensen-Shannon 距离是因为它能够衡量具有概率为 0 的类别的分布之间的距离。

We calculated the Jensen-Shannon distance using the following formula:

我们使用以下公式计算了 Jensen-Shannon 距离：

$$JSD(P||Q) = \sqrt{\frac{D_{KL}(P||M) + D_{KL}(Q||M)}{2}} \quad (1)$$

where P and Q is a probability distribution, M is $\frac{1}{2}(P + Q)$ (pointwise mean), and $D_{KL}(P||M)$ denotes the Kullback-Leibler divergence from M to P, defined as:

其中 P 和 Q 是概率分布，M 是 $\frac{1}{2}(P + Q)$ （逐点均值），而 $D_{KL}(P||M)$ 表示从 M 到 P 的 Kullback-Leibler 散度，定义为：

$$D_{KL}(P||M) = \sum_d P(d) \log \frac{P(d)}{M(d)} \quad (2)$$

for each d in demonstratives of the target language.

对于目标语言的指示代词中的每一个 d 。

We compute the Jensen-Shannon distance between the distribution of demonstrative usage by the VLM and that by humans for each distance to the target, and compute a representative value for the model by averaging these distances across target distances.

我们计算 VLM 的指示性用法分布与人类的指示性用法分布之间的 Jensen–Shannon 距离，针对每个目标距离，然后通过对这些目标距离上的距离进行平均，计算模型的代表性值。

Before the evaluation, we eliminated the inappropriate results that do not align with the format “[Demonstrative] [Color] [Shape]” and made some mistakes on colors and shapes, because VLMs fail to understand our instructions or to recognize the indicated object in those trials. As colors and shapes can be expressed in many ways (e.g. “cross” and “plus” for a cross shape), we permitted those varieties of expressions during the validation. The

在评估之前，我们去除了不符合格式 “[Demonstrative] [Color] [Shape]” 的不恰当结果，并且在颜色和形状上犯了一些错误，因为视觉语言模型（VLMs）未能理解我们的指令或在这些试验中未能识别出所指示的对象。由于颜色和形状可以用多种方式表达（例如，“cross” 和 “plus” 表示十字形状），我们在验证过程中允许这些表达方式的多样性。

judgment to determine whether the output corresponds to the synonym of the gold answer was performed manually by the author.

判断输出是否与黄金答案的同义词相符的判断工作是由作者手动完成的。

5 Results and Discussion

5 结果与讨论

5.1 Results and Analysis

5.1 结果与分析

Models	Japanese	Korean	English	Chinese
Gemma 3 4B	12	18	34	12
Gemma 3 12B	18	19	45	28
Qwen3-VL 8B	44	38	50	48
Qwen3-VL 32B	22	26	49	44

模型	日语	韩语	中文	中文
Gemma 3 4B	12	18	34	12
Gemma 3 12B	18	19	45	28
Qwen3-VL 8B	44	38	50	48
Qwen3-VL 32B	22	26	49	44

Table 1: The number of images in which the color and shape of the object are precisely recognized by the model (out of 60 images).

表 1: 模型在 60 张图像中准确识别出物体的颜色和形状的图像数量。

The number of images whose color and shape the model could precisely recognize is presented in Table 1, and the probability distributions of demonstratives for each VLM per language and distance settings are shown in Figure 2. The probability distributions shown in Figure 2 are calculated based only on the cases where images were correctly recognized by models, as presented

in Table 1. Each row of Figure 2 corresponds to an experimental setting and presents the average probability distribution of demonstrative for that setting. The bottom row is the result of the human investigations reported in Coventry et al. (2023). Green, orange, and light blue represent the probabilities of proximal, medial, and distal demonstratives, respectively.

模型能够精确识别颜色和形状的图像数量在表 1 中呈现，每种语言和距离设置下各 VLM 的指示代词的概率分布如图 2 所示。图 2 中显示的概率分布仅基于模型正确识别图像的情况进行计算，如表 1 所示。图 2 中的每一行对应一个实验设置，并展示了该设置下指示代词的平均概率分布。最后一行是 Coventry 等人 (2023) 报告的人类研究结果。绿色、橙色和浅蓝色分别代表近指、中指和远指指示代词的概率。

Across models, a general trend is to avoid distal demonstratives in languages with three demonstratives, regardless of the model. Although the proportion of distal demonstratives increased as the distance from the object increased for humans, the proportions of distal demonstratives in Korean and Japanese, which have three demonstratives, are consistently under 5% except for the case of Qwen3-VL 32B for Japanese language. Due to unbalanced distributions, models often fail to select the same demonstratives as humans do, especially in Japanese and Korean.

在不同模型中，普遍存在一种趋势，即在具有三个指示代词的语言中避免使用远指指示代词，无论模型如何。尽管对于人类来说，随着与物体的距离增加，远指指示代词的比例会上升，但韩语和日语（这两种语言都有三个指示代词）中远指指示代词的比例，除了日语在 Qwen3-VL 32B 模型中的情况外，通常都低于 5%。由于分布不平衡，模型经常无法像人类一样选择相同的指示代词，尤其是在日语和韩语中。

Regarding the trends for each model, Gemma 3 exhibits similar probability distributions across distances. Although the proportion of proximal demonstratives decreases and that of distal demonstratives increases as the distance increases for humans, changes in the probability distribution of both Gemma 3 4B and Gemma 3 12B across different distances are smaller than the results of humans. The Qwen3-VL series does not show such irrelevance to distance, but it fails to capture the human-like probability distribution.

关于每个模型的趋势，Gemma 3 在不同距离上的概率分布表现出相似性。虽然对于人类来说，随着距离的增加，近指代词的比例下降而远指代词的比例上升，但 Gemma 3 4B 和 Gemma 3 12B 在不同距离下的概率分布变化要小于人类的结果。Qwen3-VL 系列没有表现出与距离无关的特性，但未能捕捉到类似人类的概率分布。

Table 1 presents that VLMs could not recognize the figures painted on the object and the content of the prompt in some experimental settings. This phenomenon is particularly prominent in Gemma 3 4B, which exhibits biased distributions in demonstrative usage. In contrast, Qwen3-VL achieves higher recognition accuracies than Gemma 3 models across all languages. This may be caused by differences in the pre-training dataset. While the pre-training dataset of Gemma 3 is not disclosed, that of Qwen3-VL includes tasks of visual question-answering with camera-captured images (Bai et al., 2025). The fact that Qwen3-VL was pre-trained with similar tasks to ours may cause the more accurate recognition of the object in Qwen3-VL. We can also observe that with Qwen3-VL 8B and 32B, English and Chinese scores of accurate recognition are higher than Japanese and Korean ones. This can also be explained by the fact that it was pre-trained with data in which English and Chinese account for a large proportion (Bai et al., 2025).

表 1 显示，在某些实验设置下，VLMs 无法识别物体上的绘画图像以及提示内容。这种现象在 Gemma 3 4B 中尤为明显，其在指示性使用中表现出分布偏倚。相比之下，Qwen3-VL 在所有语言上的识别准确率均高于 Gemma 3 模型。这可能是由于预训练数据集的差异所致。虽然 Gemma 3 的预训练数据集未公开，但 Qwen3-VL 的预训练数据集包括使用摄像头拍摄图像的视觉问答任务 (Bai et al., 2025)。Qwen3-VL 与我们的任务具有相似的预训练任务，这可能导致其对物体的识别更加准确。我们还可以观察到，使用 Qwen3-VL 8B 和 32B 时，英语和中文的准确识别得分高于日语和韩语的得分。这也可解释为，其预训练数据中英语和中文占比较大 (Bai et al., 2025)。

It is also observable that while human preference for demonstratives varies according to distance, VLMs do not exhibit such sensitivity. For instance, in Japanese, the most frequently selected demonstratives for humans shifted from proximal kono at 0.25 m to medial sono at 1.50 m, and finally to distal ano at 2.75 m. But Qwen3-VL 32B chose the proximal demonstrative at the highest proportion regardless of distance, and the other three models chose the medial at the

highest proportion regardless of distance. This lack of the shift of demonstratives by distance may be a feature of VLMs.

也可以观察到，尽管人类对指示代词的偏好会根据距离而变化，但视觉语言模型（VLMs）并不表现出这种敏感性。例如，在日语中，人类最常选择的指示代词会随着距离的变化而变化：从 0.25 米处的近指代词“kono”变为 1.50 米处的中指代词“sono”，最后变为 2.75 米处的远指代词“ano”。但 Qwen3-VL 32B 无论距离如何，都以最高比例选择近指代词，而其他三个模型则无论距离如何，都以最高比例选择中指代词。这种指示代词不随距离变化的现象可能是 VLMs 的一个特征。

5.2 Distance from Human Distribution

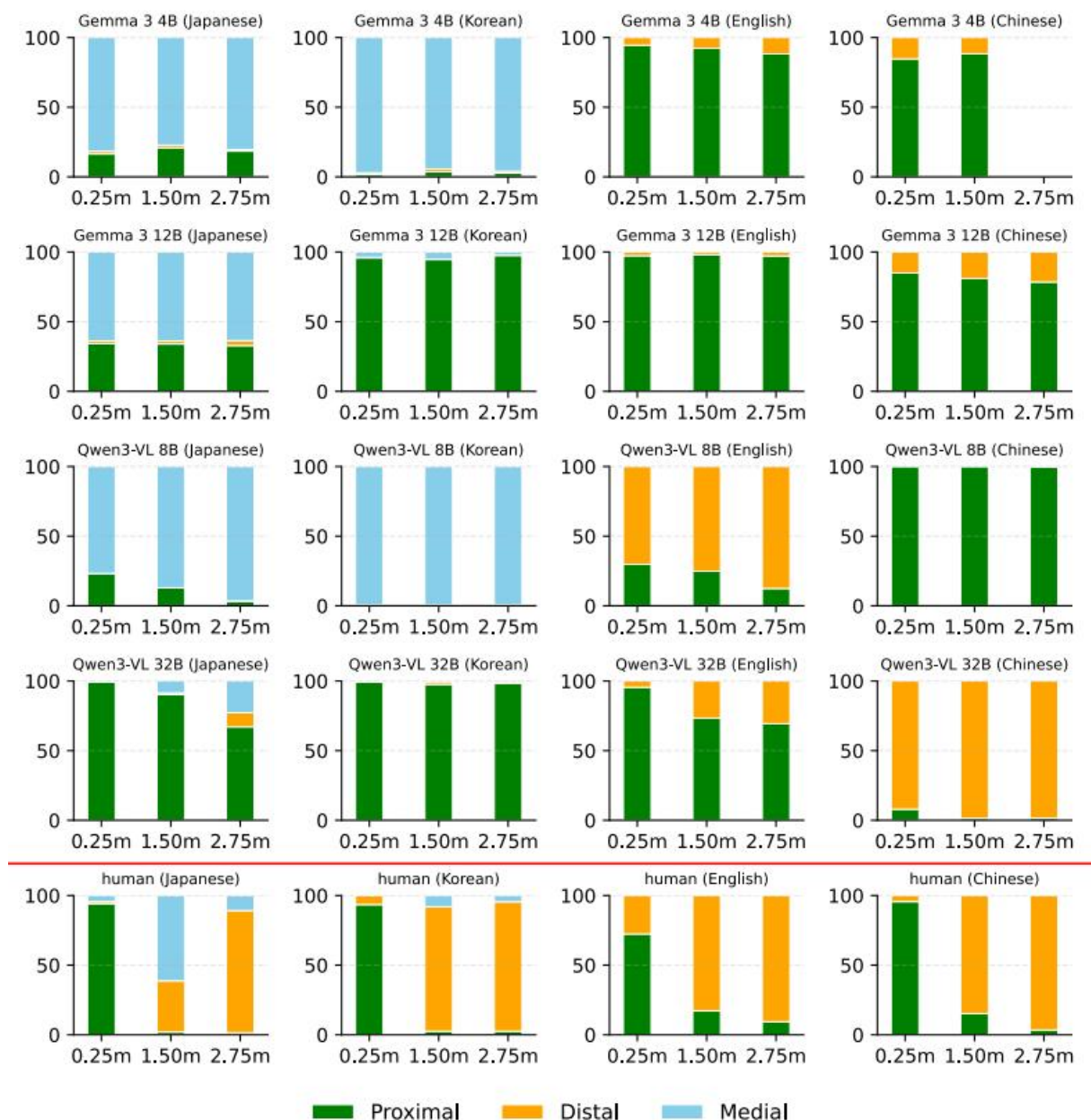
5.2 与人群分布的距离

Jensen-Shannon distances between the outputs of models and humans’ results of Coventry et al. (2023) are presented in Table 2. For comparison, the Jensen-Shannon distribution between the human distribution and the uniform distribution, in which every demonstrative is chosen with the same probability, is provided. Diagonal lines in the table indicate that we cannot calculate the distance due to insufficient data resulting from a violation of the output format.

Coventry 等人（2023）提出的模型输出与人类结果之间的 Jensen-Shannon 距离在表 2 中给出。为了进行比较，还提供了人类分布与均匀分布之间的 Jensen-Shannon 分布，其中每个示例被选择的概率相同。表中的对角线表明，由于输出格式违规导致数据不足，我们无法计算距离。

Table 2 shows Gemma 3 4B performs more differently from human distribution than uniform distribution in any language. In contrast, Qwen3-

表 2 显示，Gemma 3 4B 在任何语言中都比均匀分布更不同于人类分布。相比之下，Qwen3-



Proximal Distal Medial

近端远端内侧

Figure 2: Probability distributions across distances for each experimental setting. The red horizontal line separates results for VLMs (above) from those for humans (below). Columns represent different languages, while rows above the red horizontal line correspond to specific VLMs. Human results in the bottom row were calculated based on the results in Coventry et al. (2023).

图 2：每个实验设置下，距离的分布概率。红色水平线将 VLMs（上方）的结果与人类（下方）的结果分隔开。列代表不同的语言，而红色水平线上方的行对应特定的 VLMs。底部行中的人类结果是基于 Coventry 等人（2023）的研究结果计算得出的。

VL 32B shows a similar distribution to humans in Japanese and Chinese, therefore, we can observe that Qwen3-VL 32B uses demonstratives in a relatively close way to humans.

VL 32B 在日语和汉语中表现出与人类相似的分布情况，因此，我们可以观察到 Qwen3-VL 32B 在使用指示代词方面与人类较为接近。

We can also observe that the average distance in English is consistently smaller than that in Korean. This can be explained in two ways. First, Korean is a relatively low-resource language compared to English, and the shortage of training resources for Korean demonstratives may lead to low fidelity to humans’ distribution. Second, the probability distribution of the use of demonstratives in English is closer to a uniform distribution than that in Korean. Thus, it is a relatively easier task to answer in a closer distribution in English than in Korean.

我们还可以观察到，英语中的平均距离始终小于韩语中的平均距离。这可以从两个方面来解释。首先，与英语相比，韩语是一种资源相对较少的语言，韩语指示代词的训练资源不足可能导致其与人类分布的保真度较低。其次，英语中指示代词的使用概率分布比韩语更接近均匀分布。因此，在英语中回答一个更接近的分布任务比在韩语中更容易。

These results suggest that the lack of general object recognition ability in VLMs may cause biased demonstrative usage, even when color and shape are accurately recognized at evaluation time. To further analyze the relationship between model object recognition ability and demonstrative usage, we calculated the correlation between image recognition ability and the distance of the probability distribution across 15 experimental settings, encompassing

这些结果表明，VLMs 缺乏一般的物体识别能力可能会导致演示性用法出现偏差，即使在评估时颜色和形状被准确识别的情况下也是如此。为了进一步分析模型物体识别能力与演示性用法之间的关系，我们计算了图像识别能力与概率分布距离之间的相关性，涵盖 15 种实验设置。

various combinations of languages and model architectures. The analysis yielded a Pearson correlation coefficient of $r = -0.40$, indicating a modest negative trend between the two metrics. This result suggests that models with higher recognition proficiency tended to exhibit demonstrative usage distributions more similar to those of humans.

各种语言和模型架构的组合。分析得出皮尔逊相关系数为 $r = -0.40$ ，表明这两个指标之间存在适度的负趋势。这一结果表明，识别能力较高的模型在演示性使用分布上更接近人类的分布。

6 Conclusion

6 结论

In this paper, we developed a multilingual benchmark to assess the extent to which VLMs can use spatial deictic expressions in a manner similar to humans based on the memory game paradigm. Using our benchmark, we investigated the probability distributions of demonstratives by object distance.

在本文中，我们开发了一个多语言基准，以评估视觉语言模型（VLMs）在基于记忆游戏范式的情况下，使用空间指示表达方式的程度，这种程度类似于人类。通过我们的基准，我们研究了指示代词的概率分布，这些分布基于物体之间的距离。

We discovered that all tested models fail to reproduce humans’ probability distribution for demonstratives as a function of the absolute distance from the viewpoint to the target object. We also observed that, unlike humans, some VLMs tend not to adjust their probability distributions based on distance to the referent. These results suggest that current VLMs have significant room for improvement in their use of demonstratives in a human-like

我们发现所有测试的模型都无法再现人类在指示词的概率分布上，其与视角到目标物体的绝对距离之间的关系。我们还观察到，与人类不同，一些视觉语言模型（VLMs）倾向于不根据指代对象的距离来调整其概率分布。这些结果表明，当前的 VLMs 在以类似人类的方式使用指示词方面还有显著的改进空间。

Models	Distance(m)	Japanese	Korean	English	Chinese
Gemma 3 4B	0.25	0.7227	0.9554	0.2615	0.1597
1.50	0.4591	0.8349	0.6830	0.6571	

2.75	0.8252	0.8880	0.7174	\	
Average	0.6080	0.8705	0.5041	\	
Gemma 3 12B	0.25	0.5732	0.2067	0.3167	0.1557
1.50	0.5104	0.9173	0.7641	0.5839	
2.75	0.7904	0.9367	0.8252	0.7018	
Average	0.5815	0.7313	0.6326	0.5063	
Qwen3-VL 8B	0.25	0.6749	0.9849	0.3677	0.1495
1.50	0.4821	0.8771	0.0799	0.8221	
2.75	0.8510	0.9143	0.0413	0.9437	
Average	0.6648	0.9200	0.1539	0.6499	
Qwen3-VL 32B	0.25	0.1294	0.1638	0.2799	0.8175
1.50	0.8424	0.9130	0.4940	0.2277	
2.75	0.7489	0.9247	0.5462	0.0510	
Average	0.5055	0.7446	0.4444	0.3725	
Uniform distribution	0.25	0.5740	0.5869	0.1962	0.4626
1.50	0.3907	0.5186	0.2998	0.3216	
2.75	0.5149	0.5551	0.3912	0.4791	
Average	0.4932	0.5535	0.2957	0.4211	

模型	距离 (米)	日语	韩语	中文	中文
Gemma 3 4B	0.25	0.7227	0.9554	0.2615	0.1597
1.50	0.4591	0.8349	0.6830	0.6571	
2.75	0.8252	0.8880	0.7174	\	
平均值	0.6080	0.8705	0.5041	\	
Gemma 3 12B	0.25	0.5732	0.2067	0.3167	0.1557
1.50	0.5104	0.9173	0.7641	0.5839	
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Qwen3-VL 32B	0.25	0.1294	0.1638	0.2799	0.8175
1.50	0.8424	0.9130	0.4940	0.2277	
2.75	0.7489	0.9247	0.5462	0.0510	
平均值	0.5055	0.7446	0.4444	0.3725	
均匀分布	0.25	0.5740	0.5869	0.1962	0.4626
1.50	0.3907	0.5186	0.2998	0.3216	
2.75	0.5149	0.5551	0.3912	0.4791	
平均值	0.4932	0.5535	0.2957	0.4211	

Table 2: Distance from human distribution of demonstrative use per language.

表 2: 各语言中指示用法的人类分布距离。

way. We also clarified the difference in performance among languages. In particular, all tested models show weaknesses in Korean language. We also found that VLMs seldom use distal demonstratives in Japanese and Korean.

方式。我们还澄清了不同语言在性能上的差异。特别是所有测试的模型在韩语方面都表现出弱点。我们还发现，视觉语言模型（VLMs）在日语和韩语中很少使用远指代词。

The results described above suggest that there are differences between VLMs and humans in handling spatial deictic expression, which is one of the fundamental expressions for spatial understanding. We believe that our benchmark serves as a new testbed for evaluating the spatial reasoning abilities of VLMs across languages.

上述结果表明，VLMs 在处理空间指示表达方面与人类存在差异，而空间指示表达是空间理解的基本表达之一。我们认为，我们的基准测试为评估 VLMs 在不同语言中的空间推理能力提供了一个新的测试平台。

7 Limitations

7 局限性

Though this benchmark introduces a novel methodology to evaluate the ability to use spatial deictic expressions, it is subject to several limitations. First, the models cannot recognize a number of pictures, resulting in a constrained number of samples for evaluation. Furthermore, the scope of our experimental setup is currently restricted. Since our evaluation relies on a single prompt, it is difficult to rigorously determine whether the observed trends reflect the models' intrinsic capabilities or are artifacts of the prompt's phrasing. The generalizability of these results is also constrained by the limited size of the dataset used and the limited number of languages and model architectures evaluated in this study. Increasing the dataset size in future studies is required to mitigate potential biases and ensure a more robust assessment of the results. In addition, since we needed to use a controlled prompt setting to ensure the VLM outputs satisfy the expected format (demonstrative, color, and shape), the variety of usage of demonstratives in natural situations is not sufficiently reflected. Establishing more naturalistic settings remains as future work. Finally, the evaluation process involved human judgment to verify model outputs. It introduces an element of subjectivity and limits the benchmark's scalability.

尽管这个基准测试引入了一种新颖的方法来评估使用空间指示表达的能力，但它存在一些限制。首先，模型无法识别一些图片，导致评估样本数量受限。此外，我们的实验设置范围目前受到限制。由于我们的评估依赖于单一提示，因此很难严格确定观察到的趋势是反映了模型的内在能力，还是提示语措辞的产物。这些结果的可推广性也受到所用数据集规模有限以及本研究中评估的语言和模型架构种类有限的限制。未来的研究需要增加数据集的规模，以减轻潜在的偏见，并确保对结果进行更稳健的评估。此外，由于我们需要使用受控提示设置来确保视觉语言模型的输出符合预期格式（指示性、颜色和形状），因此在自然情境中指示性表达的使用多样性并未得到充分体现。建立更加自然的情境仍然是未来的工作。最后，评估过程涉及人类判断以验证模型输出，这引入了主观性因素，并限制了基准测试的可扩展性。

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